Wuyang Chen

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Education

- **City University of Hong Kong** M.Sc. in Artificial Intelligence GPA: 3.70/4.30 *09/2025 - 10/2026*
  - **Core Courses:** Vision and Image (A), Intelligent Systems (A), Artificial Intelligence (A), Topics in Autonomous Driving (A), Dynamic Programming and Reinforcement Learning (A-)
- **Beijing Jiaotong University** B.Eng. in Computer Science and Technology GPA: 86.6/100 *09/2021 - 06/2025*
  - **Core Courses:** Mathematical Analysis I & II (100 & 99), University Physics I & II (97 & 98), Advanced Algebra (93), Deep Learning (90), C++ (97), Data Structures (95), Python (A)
  - **Honors:** National Scholarship (Top 1.5%, 12/2022), First-Class Academic Scholarship (Top 3%, 12/2022), "Triple-A Student" (Top about 10%, 12/2022)

Publications

- **Wuyang Chen**, **Wenqi Wu**, and **Jun Wang**. "A Cascaded Planner-Governor Approach to Path Planning of Autonomous Underwater Vehicles with Obstacle Avoidance" *Submitted to IEEE Transactions on Intelligent Vehicles (T-IV)*. Under Review.

Research

- **A Cascaded Planner-Governor Approach to Path Planning of Autonomous Underwater Vehicles with Obstacle Avoidance** *01/2026 - Present*
  Tech: Optimization, Control Engineering, MPC, Hydrodynamics, ROS Noetic, MuJoCo [Github]
  - **Cascaded Architecture Design:** Proposed a global-local path planning approach featuring a cascaded planner-governor to guarantee both the moving-obstacle avoidance and actuator protection. Used an nonlinear ESO to estimate the hydrodynamics disturbance based on the 3-DOF Fossen model.
  - **Improved Receding Horizon Planner (IRHP):** Designed an IRHP featuring a projection-based switching mechanism to prevent minimum deadlocks, an improved polynomial barrier function to retain continuous optimization gradients, and a multi-objective cost function to weight each element.
  - **Jerk-Layer-Limited Reference Governor (JLLRG):** Designed a JLLRG to smooth the commands and protect the actuator. Took yaw torque, yaw acceleration, and yaw jerk limits into account to address command oscillations and actuator limit violations.
  - **Closed-Loop Sim-to-Real Testbed:** Developed a testbed integrating ROS Noetic for distributed node communication (IRHP, JLLRG, ESO) and MuJoCo for hydrodynamics simulation, ready for hardware experiments.
  - **Comparative Simulations:** Evaluated the proposed approach against state-of-the-art existing approaches under extreme scenarios with U-shaped obstacles, misaligned slits, and moving obstacles.
  - **Achievements:** Achieved an approximately 81% decrease in yaw torque command oscillation and 20% decrease in energy consumption compared with existing approaches.
- **Multi-Object Hybrid Tracking and Future Dynamic Prediction System** *02/2026 - 05/2026*
  Tech: Autonomous Driving, LSTM, Voxelization, Deep Learning + Physics Modeling [Github]
  - **Perception & Tracking:** Processed real-world high-dimensional LiDAR point cloud data (~35k points/frame). Designed a 3D object detection pipeline based on 0.2m high-precision Voxelization, RANSAC, and DBSCAN clustering, reducing perception false positives from 10688 to 6800.
  - **Sequential Modeling:** Addressed nonlinear turn drift for agents in complex environments by building a high-precision kinematic baseline based on constant turn rate and velocity. Utilized an LSTM architecture to encode high-dimensional history trajectories to capture nonlinear evolution patterns.
  - **Achievements:** Dropped the overall average L2 prediction error from 3.74m to 2.64m. Reduced the L2 error from 4.43m to 3.25m under a limited prediction window of 5.5s. Overcame the tangential divergence effect of traditional models in long-term prediction.

INTERNSHIPS

- **Southeast University** Research Intern *04/2026 - 06/2026*
  Tech: Unmanned Aerial Vehicle, Extended Kalman Filter (EKF), Relative Localization, Factor Graph, ROS Noetic
  - **Literature Analysis:** Conducted literature research on relative localization of unmanned aerial vehicle, focusing on EKF, factor graph, and multi-sensor fusion.
  - **Algorithm Optimization:** Addressed the unknown cross-correlation issue inherent in EKF. Integrated the covariance intersection algorithm to fuse state estimates via a pessimistic convex combination, purposefully inflating the covariance matrix to prevent filter overconfidence and ensure robust estimation.
  - **High-Fidelity Simulation Development:** Engineered a 3D flight co-simulation platform by Python and ROS, establishing a reliable closed-loop testing environment for upcoming real-world experiments.
- **Institute of Software, Chinese Academy of Sciences** Research Intern *04/2024 - 11/2024*
  - Participated in the Alzheimer's disease machine recognition, specifically trained YOLOv5 networks.
  - Developed a dataset annotation software and constructed data processing workflows.
  - Developed wrote handwriting recognition algorithms, involving pixel aggregation, linkage models, and branch-and-bound strategies.
- **Tencent Technology (Shenzhen) Co., Ltd.** Front-end Development Intern *08/2025 - 09/2025*
  Tech: Flutter Architecture, Routing Navigation, Protocol Communication, Android & IOS development
- **4399 Network Co., Ltd.** Client Development Intern *04/2025 - 05/2025*
  Tech: Unity Engine, MVC Architecture, JavaScript & TypeScript

ENGLISH

- **IELTS 6.5** 04/2024
- **GRE 319 (math with full marks) +3.5 (writing)** 12/2024

SKILLS

- **Programming & Tools:** Python, C/C++/C#, Assembly, MATLAB, Linux, Git, Docker
- **Frameworks & Simulation:** PyTorch, ROS Noetic, MuJoCo, Unity
- **Control & Localization:** MPC, ESO, EKF and its variations(e.g., Iterated EKF, Invariant EKF)
- **Machine Learning:** Reinforcement Learning (e.g., Deep Q-Network, Actor-Critic), Computer Vision (e.g., U-Net, YOLO), Sequence Modeling (e.g., LSTM, Transformer)